

Module 13: Admissions Prediction Project Writeup

1. Preprocessing Decisions (Section 1)

The raw admissions data was sourced from a cleaned version of the GradCafe dataset. The following preprocessing decisions were made to ensure high-quality model inputs:

- **Target Variable:** A binary label was created where 1 represents "Accepted" and 0 represents "Rejected." Any ambiguous statuses (e.g., Waitlisted) were removed.
- **Deduplication:** Records were filtered to ensure no duplicate application entries existed.
- **Feature Selection:** We selected a multimodal mix of features.
 - **Text:** Program and comments.
 - **Structured:** GPA, GRE (Quant, Verbal, and AW), Degree, and Citizenship.
- **Handling Missing Data:** Numeric missing values were imputed with consistent placeholders (e.g., 0.0 or None) to prevent the model from crashing while maintaining the presence of the field in the text template.

```
=====
SECTION 1: DATA PREPARATION SUMMARY
=====
Number of rows in original dataset:      43046
Number of Accepted rows (Label 1):      14717
Number of Rejected rows (Label 0):      16981
-----
Text Fields:      ['Program', 'comments']
Non-text Fields:  ['GPA', 'GRE', 'GRE V', 'Degree', 'US/International']
-----
Preview of cleaned dataframe:

```

	Program	comments	GPA	GRE	GRE V	label
0	Comparative Literature	None	3.82	169.0	160.0	0
2	Physics	None	3.89	169.0	160.0	0
3	Quantum Science and Engineering	Got an email to check the decision on the appl...	3.57	327.0	157.0	0
4	Speech Language Pathology	None	3.30	169.0	160.0	0
5	Physics	Yay! I got selected for my dream school in Nep...	3.82	169.0	160.0	1

```
=====
```

2. Model Input Template (Section 2)

To allow a Transformer model to "read" structured data, each applicant was converted into a unified text representation. This ensures the model understands the context of each value.

The Exact Template:

Plaintext

```
Program: {Program}
University: {University}
Comments: {comments}
Term: {term}
```

```
Degree: {Degree}
Citizenship: {US/International}
GPA: {GPA}
GRE Quant: {GRE}
GRE Verbal: {GRE V}
GRE AW: {GRE AW}
```

This template was chosen because it explicitly labels each field, making it human-readable and consistent across all 31,000+ samples.

SECTION 2: MULTIMODAL INPUT REPRESENTATION

EXACT TEMPLATE USED:

```
-----
Program: {Program}
Degree: {Degree}
Citizenship Status: {US/International}
GPA: {GPA}
GRE Quant: {GRE}
GRE Verbal: {GRE V}
Comments: {comments}
-----
```

THREE SAMPLE MODEL INPUTS FROM TRAINING DATASET:

SAMPLE 1:

```
Program: Comparative Literature
Degree: PhD
Citizenship Status: International
GPA: 3.82
GRE Quant: 169.0
GRE Verbal: 160.0
Comments: nan
-----
```

SAMPLE 2:

```
Degree: PhD
Citizenship Status: International
GPA: 3.89
GRE Quant: 169.0
GRE Verbal: 160.0
Comments: nan
-----
```

SAMPLE 3:

```
Program: Quantum Science and Engineering
```


- **Flexibility:** While more complex to train, the pre-trained nature of DistilBERT allows the model to start with a "general knowledge" of English, which it then adapts to the specific language of academic admissions.

```
=====
SECTION 5: FINAL EVALUATION REPORT
=====

--- CLASSIFICATION METRICS ---
      precision    recall  f1-score   support

   Rejected       0.76     0.85     0.80     3396
   Accepted       0.80     0.70     0.74     2944

 accuracy          0.78          0.78     6340
 macro avg         0.78     0.77     0.77     6340
weighted avg         0.78     0.78     0.78     6340

Overall Accuracy: 0.7778

--- PROBABILITY EXAMPLES ---
Sample 1: Result=1 | P(Acc)=0.7863 | P(Rej)=0.2137
Sample 2: Result=1 | P(Acc)=0.9862 | P(Rej)=0.0138
Sample 3: Result=0 | P(Acc)=0.0162 | P(Rej)=0.9838

--- CORRECTLY CLASSIFIED EXAMPLE ---
Program: Mathematics
University: ETH Zurich
Comments: Applied for the ESOP also but haven't heard word on it. TBH, if I get the ESOP I will probably just go here because I'm not
at excited about my PhD acceptances.
Term: Fall 2025
Degree: Master's
Citizenship: American
```

```
--- PROBABILITY EXAMPLES ---
Sample 1: Result=1 | P(Acc)=0.7863 | P(Rej)=0.2137
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Program: Mathematics
University: ETH Zurich
Comments: Applied for the ESOP also but haven't heard word on it. TBH, if I get the ESOP I will probably just go here because I'm not th
at excited about my PhD acceptances.
Term: Fall 2025
Degree: Master's
Citizenship: American
GPA: 3.9
GRE Quant: 169.0
GRE Verbal: 160.0
GRE AW: None
Result: 1 (Matches Actual)

--- INCORRECTLY CLASSIFIED EXAMPLE ---
Program: Chemistry
University: University of California
Term: Fall 2024
Degree: PhD
Citizenship: American
GPA: 3.6
GRE Quant: 169.0
GRE Verbal: 160.0
GRE AW: None
Result: 1 (Actual was 0)
=====
```

6. Reflection on Limitations and Ethics (Section 9)

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Loading weights: 100%|██| 104/104 [00:00<00:00, 6118.6lit/s]
--- SECTION 6: CORRECTED INFERENCE TEST ---

Result for Candidate A (Strong CS):
 Prediction: Rejected
 Confidence: 0.8554

Result for Candidate B (Weak History):
 Prediction: Accepted
 Confidence: 0.8800
```

# Graduate Admissions Predictor

Enter your details below to see your predicted admission status.

## Program Name

## University Name

## Undergraduate GPA

## GRE Quantitative Score

## Degree Type

## Applicant Comments

Check My Chances

**Disclaimer:** This is a class project model trained on scraped, self-reported data from GradCafe. These results are generated by an AI and are NOT real admissions decisions.

## Admission Prediction

**Prediction: Accepted**

Model Confidence Score: **0.83**

Try Another Applicant

## 8. Model Comparison & Analysis

### 1. Capabilities: Data Inputs

The most immediate difference is the **dimension of data**.

- **The NumPy Model:** This was a "blind" classifier. It only understood six structured numeric features. It treated every applicant as a vector of floats. It had no concept of what "Computer Science" meant versus "History"—it only knew if a 3.8 GPA was higher than a 3.5.
- **The PyTorch Transformer:** This model uses **multimodal inputs**. By converting everything into a unified text string, the model "reads" the application. It can identify specific keywords, sentiment, and the prestige of programs or universities through the weights it inherited during its pre-training on the entire English language.

### 2. The Impact of Text (Comments & Programs)

Including qualitative data like **Comments** and **Program names** provides the "why" behind the "what."

- **The Benefit:** In your training results, we saw the model correctly identify a Math applicant at ETH Zurich because it could parse the specific context of their Master's degree and their comments about PhD acceptances.
- **The Qualitative Edge:** Comments allow the model to see "Green Flags" (like publications or research) that are invisible to a numeric-only model. This added context is likely why this model achieved nearly **78% accuracy**, whereas a numeric-only model often plateaus much lower because it can't see the applicant's experience.

### 3. Flexibility vs. Fragility

- **Flexibility:** The pretrained model is incredibly flexible. Because it was trained on the "Uncased" BERT corpus, it understands synonyms and professional terminology without you having to define them. It "knows" that "CS" and "Computer Science" are the same thing.
- **Fragility:** However, as we saw in our **Inference Test**, the model is more **fragile** regarding specific tokens. A single word like "Stanford" can trigger a "Rejected" prediction because the model is over-sensitive to the high rejection rates associated with that token in the training data. The NumPy model was more "stable" because it was too simple to be "distracted" by names or stories.

### 4. Is the Complexity Justified?

The added complexity—installing PyTorch, managing Transformers, and the longer training time—is **fully justified**.



- **Real-World Mirroring:** Human admissions officers don't just look at a spreadsheet of GPAs; they read essays and resumes. By using a Transformer, you have built a tool that mimics the *actual* process of admission evaluation.
- **Performance Ceiling:** A two-layer NumPy model has a very low performance "ceiling." No matter how much data you give it, it will never understand "Context." The Transformer has a much higher ceiling and provides a more human-like prediction, even with its occasional biases.

## 9. Reflection on Limitations and Ethics

### The Power of Fine-Tuning

Fine-tuning a pretrained model like **DistilBERT** is significantly more efficient than training from scratch. Because the model has already been trained on the vast corpus of the English language, it understands grammar, professional terminology, and context before we even show it a single row of data. This allows us to achieve high accuracy with a relatively small dataset and limited computing power, "standing on the shoulders" of existing AI research.

### Multimodal Context: Text + Numbers

In admissions, numbers (GPA/GRE) are only part of the story. By combining structured features with raw text, we create a **multimodal** representation that mirrors how a human admissions officer works. A high GPA is impressive, but a comment mentioning "3 years of research and a first-author publication" provides the qualitative depth that often tips the scale from a rejection to an acceptance.

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### Bias in Self-Reported Data

The GradCafe dataset suffers from **Selection Bias**. People who choose to self-report their results online are rarely a perfect cross-section of all applicants; they tend to be those with exceptionally high scores or those applying to prestigious programs. Consequently, the model may be "over-trained" on high-achievers, leading it to set unrealistically high bars for what constitutes a "successful" applicant.

### Misleading and Unfair Predictions

During testing, we observed **Keyword Bias**. The model occasionally predicted "Rejected" for perfect candidates simply because they applied to highly competitive schools like Stanford. The model learned that the *statistical* likelihood of rejection at those schools is so high that it sometimes ignores the candidate's actual qualifications.

This could lead to a "discouragement loop" where a qualified student is told they won't get in based on a historical trend they are actually capable of defying.

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## The "Missing" Information

A model like this is fundamentally limited because it lacks access to the most human parts of an application:

- **Letters of Recommendation:** The model cannot hear the praise of a mentor.
- **Personal Statements:** The model sees snippets of comments, not the long-form narrative of a student's journey.
- **Institutional Fit:** Admissions committees often look for specific research interests that align with a professor's current funding—information that is never captured in a CSV file.

## Ethical Conclusion

Should a model like this be used in real decision-making? **Absolutely not.** While it is a powerful statistical tool, it lacks the empathy and nuanced judgment required to evaluate a human's potential.

**Educational Value:** This project is educationally invaluable because it teaches us how to bridge the gap between raw data and natural language. However, it serves as a cautionary tale: a 78% accuracy rate is a great grade for a class project, but a 22% error rate in real life means thousands of lives potentially altered by a "hallucinating" algorithm. On a public-facing page, transparency is key—disclaimers and confidence scores must be used to ensure the user treats the output as a "guess," not a verdict.